

AMATH 582: Home Work 3

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This report is devoted to comparing the performance of different classifiers through a classic problem of machine learning: recognizing handwritten numbers. The dataset consists of a large number of pictures, each of which is expressed by pixels. Because the data set is too large, PCA methods are used for dimensionality reduction. After dimensionality reduction, we classify and analyze the new data and compare the results using Ridge, KNN, LDA and SVM classifiers.

Introduction and Overview

Handwritten digit data set (MNIST) is a very common data set in machine learning. This data set consists of two parts: training set and test set, in which the training set and test set contain features and labels respectively. The size of the features in the training set is $28 \times 28 \times 60000$, where 60000 represents a total of 60,000 images, and 28×28 represents the features of each image, that is, its pixels in a 28×28 graph. This also means that through these pixels, we can restore the image of the hand to write the number, and the label set records the specific number on the image. Similarly, the size of the test set feature is 784×10000 , indicating that 10,000 images are stored, and the number represented by each image has a test set label record.

The ultimate goal of machine learning is to find some classification method by studying the training set, and realize the corresponding label inference from the feature data. In general, the effectiveness of the classifier can be judged by the prediction accuracy of the cross-validation and classification methods on the test set. Also, that is what this report will accomplish.

Theoretical Background

Since the data set is too large, it is necessary to reduce the dimension of the data set in order to ensure the calculation speed. The dimensionality reduction method adopted in this report is PCA method. This is a method based on Singular Value Decomposition: $A = U \Sigma V^T$, by multiply the centered data matrix with the Left Singular Vector U , the dimension reduced dataset can be obtained as $\tilde{X} = U^T X_{centered}$. However, because the SVD method is computationally heavy, this report will use Python's built-in PCA function directly.

Based on dimensionality reduction data, some classifiers can be used. For instance, the first classifier used in this report is the Ridge classifier. It's based on

the linear regression. It introduces the L2 regularization to limit the size of the model coefficients, which means that its goal is to find a β satisfies $\hat{\beta} =$

$\min_{\beta \in \mathbb{R}^J} \frac{1}{2\sigma^2} \|A\beta - y\|^2 + \frac{\lambda}{2} \|\beta\|_2^2$, where A is the feature and y is the label. In the

classification task, it converts the target label to $\{-1, 1\}$ and makes category predictions by setting the threshold to zero.

What's more, the KNN method is also introduced. For a given input data point,

KNN computes the Euclidean distance (i.e. $\sqrt{\sum_{i=0}^n (x_i - y_i)^2}$) between this point

and every other point in the training dataset. Then, it selects the k training points that are closest to the input point and predicts the class of the input point by majority voting among its k neighbors.

Also, the LDA method is also used. It's based on the Bayes' theorem. Consider the probability density function for a data point x belonging to class c is given by:

$p(x|y = c) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_c)^T \Sigma^{-1} (x - \mu_c) \right)$, where μ_c is the mean,

vector of class c and Σ is the common covariance matrix and d is the dimensionality of the data. The posterior probability of class c given x is

$p(y = c|x) = \frac{p(x|y = c)p(y=c)}{p(x)}$. Thus, the LDA classifier assigns x to the

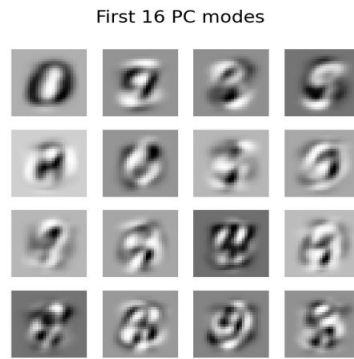
class c that maximizes the posterior probability $p(y = c|x)$.^[1]

Algorithm Implementation and Development

In this report, numpy package is used for loading and processing the data, while the matplotlib package is used for visualizing the digit images. What's more, the sklearn package is important for the report, since it's helpful to introduce the PCA method and the classifiers, such as the Ridge, KNN, LDA and SVM classifiers. Beyond that, the sklearn package also provides a convenient way to get the accuracies of the classifiers.

Computational Results

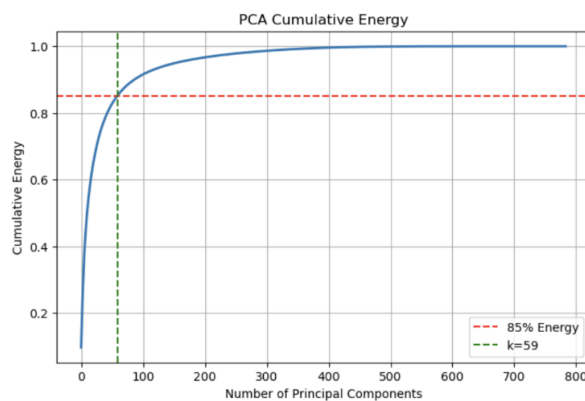
Since we've already discussed that the dimensional reduction is required for processing the large data set, the first thing we need to do is to use the PCA method. To make PCA method available, we need to reshape the dataset from $28*28*60000$ to $784*60000$. After that, the PCA method can be implemented, and here's an example of the first 16 PCA components:



The figure of the first 16 PC modes

However, such number of components seems to be not enough to express the essential information about the feature of the digit images. Thus, the next thing we need to do is to check the cumulative energy of the singular values and determine the minimum k which ensures an approximation of 85% energy.

Here's the result:



The figure of PCA cumulative energy depends on k

It's obvious that at least 59 PC mode are required. To check the conclusion, we can plot the reconstructed digit images from 59 truncated PC modes. Here are some examples:

Reconstructed digit images from 59 truncated PC modes



The first 64 reconstructed digit images from 59 truncated PC modes

Through the images, we can find that the features of the digits are kept well.

To do some basic analysis on the PCA space, we can start by doing some binary classification tests, that is, taking a set of only two numbers and training the classifier to determine which number the test data belongs to.

Firstly, we tried on the data from the sets of digit 1 and 8. The way to extract such subset is to first determine whether each image is a number 1 or 8 through a set of tags, and then cut out information about the images with numbers 1 and 8 in the feature set and the tag set, respectively.

By doing that, the shape of the sub-trainset which only contains the feature of digits 1 and 8 we got is 784×12593 , while the shape of the sub-testset is 784×2109 . Next, following the previous conclusion, we convert the sub-trainset to the 59-PCA space, and then use the converted matrix to train the Ridge classifier. Finally, we tested the effects of the Ridge classifier by cross-validation and implementing it on the converted sub-testset. The accuracy of the cross-validation is about 96.35%, and the accuracy of the prediction to the test set is 97.96%. That's a good result since the accuracies are both high enough, it shows that the classifier not only is not affected by different data divisions, but also has good adaptability to new data sets. What's more, the accuracy of the cross validation seems to be close to the accuracy of testing prediction, which indicates that there was no overfitting or underfitting. Thus, the Ridge classifier has a good effect on the binary classification missions.

What's more, we can also do the same test on the sub-set, which only contains 3 and 8 or 2 and 7 separately. Using the same selecting function, we can get the sub-trainset which only contains the feature of digits 3 and 8 with the shape of 784×11982 and the sub-trainset which only contains the feature of digits 3 and 8 with the shape of 784×1984 . Next, by using the same way to convert the sub-trainset to the 59-PCA space and train the Ridge classifier. To test the effect of the classifier to distinguish 3 from 8, we note that the accuracy of cross-validation is 95.94% and the accuracy of testing is 96.37%. It's still a good result, although the accuracies are a little bit lower than the accuracies to distinguish between 1 and 8. That's probably because 3 and 8 look more like each other and their principal component similarity is higher, which makes it harder to compute the Maximum A Posteriori(MAP).

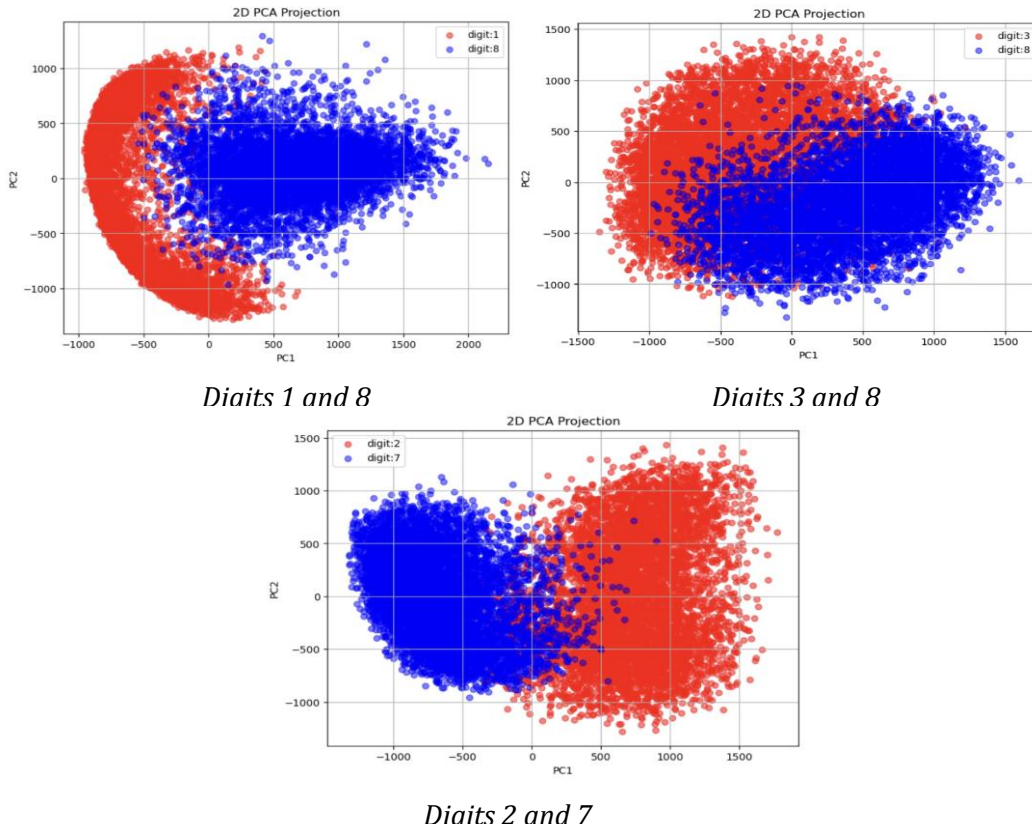
Similarly, if the same test is implemented to the digits 2 and 7, we can get the sub-trainset with the shape of 784×12223 and the sub-trainset which only contains the feature of digits 3 and 8 with the shape of 784×2060 . Again for the data on the 59-PCA space, the accuracy of cross-validation by the Ridge classifier is 97.33% and the accuracy of testing by the Ridge classifier is 98.03%. It's very close to the result of the accuracy to classify digits 1 and 8, which indicates that it has a good result as well and the difficulty to distinguish 1 from 8 and 2 from 7.

	Cross-validation accuracy	Testing accuracy
1 and 8	96.35%	97.96%

3 and 8	95.94%	96.37%
2 and 7	97.33%	98.07%

The table of the cross-validation accuracy and testing accuracy of different subsets

Here are some figures of the digits in 2-PCA space, which can partly support our guess for the reasons of the difference and similarity of the classifiers results:



The figures of digits from subtrain-sets in 2-PCA space

Finally, we can also try it on the whole data set. However, since now we have 10 kinds of labels for the data, the classification is no longer binary. Therefore, the labels may be turned into some other forms to satisfy the classifier. For instance, the label {0} can be expressed as {1,0,0,0,0,0,0,0,0} in the Ridge classifier. Moreover, we can try different classifiers to test their abilities for classification. To ensure the speed of calculation, the PCA truncation is still required. Here are some results:

	Cross-validation accuracy	Testing accuracy
The Ridge Classifier	84.38%	85.59%
The KNN Classifier	97.46%	97.57%
The LDA Classifier	86.50%	87.49%
The SVM Classifier	87.45%	88.52%

The table of the cross-validation accuracy and testing accuracy from different classifiers

From the results, it's obvious that for all the classifiers, the accuracy of the cross validation seems to be close to the accuracy of testing prediction, which indicates that there was no overfitting or underfitting anywhere. What's more, through the value of the accuracy, we can make a conclusion that the KNN Classifier has the highest accuracies and is the best multi-class classifier for the mission to distinguish figures of all digits. The results of the other classifiers are similar, which shows that the ability of the SVM method to process the multi-class classification is inferior to the KNN method but a little bit superior to the Ridge method and the LDA method.

Summary and Conclusions

To draw a conclusion, when we meet some large data sets, PCA is a good way to reduce the dimension and keep the calculation pace. The results of the PCA method can be easily checked by the cumulative energy and the reconstructed matrix. After dimension reduction, classifier can be used. For the binary classification mission, for instance, the mission to distinguish a digit figure from another, the Ridge classifier is effective, since it can provide a high cross-validation accuracy and testing accuracy without any overfit or underfit. However, the total accuracy is still related to how similar the features of the digits are. For example, the accuracy to classify 3 and 8 is lower than the accuracy to classify 1 and 8 or 2 and 7, since 3 and 8 looks more similar. Moreover, the Ridge classifier seems not to be so effective for the multi-class classification, for instance, its accuracy to classify all the ten digits figure is not so high. Actually, compared to the Ridge classifier, the LDA classifier and the SVM classifier, the KNN classifier shows a better effect. It has a high validation accuracy and testing accuracy without any overfit or underfit for the classification of all digits.

Acknowledgements

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References

- [1] Nazemi A, Maleki A. Artificial neural network classifier in comparison with LDA and LS-SVM classifiers to recognize 52 hand postures and movements[C], 2014, 2014: 18-22.